

GIES MSBA WORKSHOP SERIES

AI, ML, and Financial Markets

Nine hands-on sessions on AI agents, machine learning, and how they're reshaping investment research — what we'll cover and how it works.

Interactive textbook: dl-msba-workshop.azurewebsites.net/intro.html

Every chapter is set at Champaign Capital Research

- A fictional independent equity research firm adopting agents and ML one careful step at a time.
- The firm, its people, its problems, and its year plan are on the firm page.
- Read it first — every chapter picks up a task from it.

Firm page: dl-msba-workshop.azurewebsites.net/the-firm.html

YOUR ROLE

You're the firm's new quantitative analyst

Your job at Champaign Capital: build the agents and ML systems that assist the firm's investment research.

WHY IT MATTERS

Every chapter hands you a real task from that job — you're doing the work, not watching demos.

THE SERIES AT A GLANCE

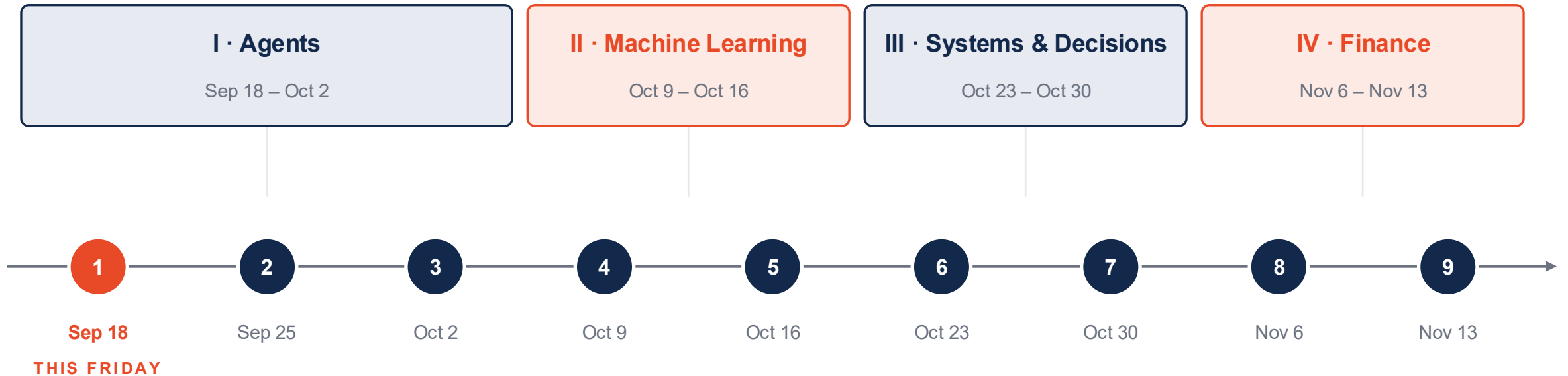
9 × 4

Nine chapters in four parts — from your first AI agent to using AI in real investment research.

Every chapter is hands-on: code you run in the browser, plus a Colab notebook.

THE JOURNEY · EVERY FRIDAY

Nine Fridays, four parts — Sep 18 to Nov 13



One chapter per session. We kick off this Friday, September 18.

First, you'll build and steer AI agents

- Ch 1 — Introduction to AI Agents: what they are, how they work.
- Ch 2 — Prompt and Context Engineering: getting reliable output.
- Ch 3 — Memory, Retrieval and RAG: giving agents your data.

Then, the ML foundations behind it all

- Ch 4 — ML Foundations I: the core ideas, from the ground up.
- Ch 5 — ML Foundations II: applying models to business problems.

Next, making it work in the real world

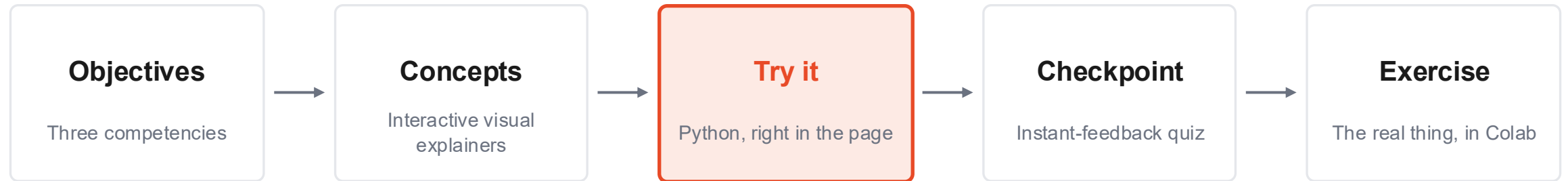
- Ch 6 — Agent Systems: from a single agent to production systems.
- Ch 7 — When to Use ML, Agents, or Neither: the judgment call.

Finally, AI meets financial markets

- Ch 8 — Financial Markets and AI as an Investment Theme.
- Ch 9 — AI for Financial and Investment Research.

HOW EACH CHAPTER WORKS

Every chapter follows the same shape — learn it once



No local setup needed — code runs in your browser, and exercises use Colab with the Lumen campus LLM service.

One thing to do before we start: the setup pages

- Read the firm page, then complete setup — about ten minutes.
- You'll get your Lumen API key working for the Colab exercises.
- Everything lives at the workshop site; bookmark it.

See you in session one.

GO DEEPER

Workshop site: dl-msba-workshop.azurewebsites.net/intro.html

Start here: firm overview + setup (~10 minutes)